AMAN KUMAR JHA

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EDUCATION

National Institute of Technology, Silchar — July 2023 – Expected June 2027
B.Tech in Electrical Engineering; CGPA: ***8.95 / 10.0*** — *Assam, India*

Jawahar Navodaya Vidyalaya, Mouli Panchkula — April 2019 – March 2022
CBSE Class 12 (PCM): ***93.2%*** | *CBSE Class 10:* ***92.2%*** — *Haryana, India*

ACHIEVEMENTS

- **Guardian** badge on LeetCode (peak **2133**, **1100+** problems solved); **Specialist** on Codeforces (max **1436**); **3 Star** on CodeChef.

EXPERIENCE

Accenture | Certificate | Source Code — May 2026 – July 2026
Advanced Application Engineering Intern – Backend Developer — *Bengaluru, India*

- **DocuMind, Multi-Document RAG Platform:** owned document ingestion and query orchestration for a **Java** / **Spring Boot** question-answering system developed by an intern team.
- Routed **4 source types** (PDF, image, text, Wikipedia) through type-specific parsers for chunking and embedding.
- Enabled asynchronous document ingestion through **Redis**-queued workers that handle parsing, embedding, and indexing.
- Skipped re-embedding exact duplicate documents using content-hash deduplication; added owner-aware access control and cascading deletion across session and global views.
- Implemented request orchestration and a planner converting questions and chat history into validated **JSON** retrieval plans; contributed to hybrid dense + keyword retrieval and cross-encoder reranking.
- *Stack:* ***Java 21, Spring Boot, PostgreSQL, Redis, Cloudinary,*** *React.*

PROJECTS

Sluice – Durable Job Queue & Scheduler — Aug 2026 – Present
Full Stack Developer | Source Code | Live — *Personal Project*

- Developed a **PostgreSQL**-backed durable queue in **Java 21** / **Spring Boot**; coordinated concurrent claims with **SELECT FOR UPDATE SKIP LOCKED**, verified through **Testcontainers** concurrency tests in a **50-test Java suite**.
- Recovered abandoned jobs using time-boxed leases, worker heartbeats, and expired-lease reclamation; supported at-least-once delivery with jittered exponential retries and dead-letter handling.
- Supported idempotent enqueueing, atomic batches, priorities, and cron schedules; respected upstream **429** responses and **Retry-After**.
- Benchmarked **1–8 workers** against a mock upstream: throughput increased from **9.2** to **44.8 jobs/sec** (**4.9x**), while p95 enqueue-to-completion latency fell from **10.2s** to **1.9s**.
- Created a **React/Vite** dashboard demonstrating **7** reliability scenarios with configurable workers and queue animations; validated with **17 frontend tests** and deployed separately on **Netlify**, **Render**, and **Neon**.
- *Stack:* ***Java 21, Spring Boot, PostgreSQL, JDBC, Flyway, React, Vite, JUnit, Testcontainers, Docker Compose, Prometheus, Grafana.***

DevTinder – Web & Android Developer Platform with Agentic RAG — Jan 2026 – Present
Full Stack Developer – Java / Node.js / React | AI Service | Core API | Frontend | Live | Demo — *Personal Project*

- Integrated **React/Redux** with **Node.js/Express** authentication, profiles and **Socket.IO** chat, plus a **Java 25/Spring Boot** service to answer questions over uploaded resumes.
- Implemented signed **Cloudinary** uploads to keep file bytes off the API; used **Apache Tika**, overlapping chunks (**1000**-character target, **200** overlap), **Pinecone** embeddings and **MongoDB** metadata to make resumes searchable.
- Built **JSON** retrieval plans with up to **3** evaluation/rewrite rounds; fused **Pinecone** dense and **MongoDB** sparse results using **Reciprocal Rank Fusion** (k=60), with a per-document chunk cap to limit domination by a single resume.
- Moved generation to **Redis** workers and streamed **Server-Sent Events** updates to keep HTTP threads free; authenticated Node-to-Spring AI requests using **HMAC-SHA256**.
- Released a **Capacitor** Android app with **Java** plugins for Keystore-backed encrypted sessions, read-only offline chats, speech and PDF previews; added **Firebase Cloud Messaging** notifications with a **MongoDB transactional outbox**, exponential backoff, lease-based recovery and duplicate-alert suppression.
- *Stack:* ***Java 25, Spring Boot, Node.js, MongoDB, Redis, Pinecone, Apache Tika, Cloudinary, Socket.IO, React,*** *Redux, TailwindCSS,* ***Capacitor, Firebase Cloud Messaging****.*

TECHNICAL SKILLS

- **Languages:** Java, JavaScript, SQL
- **Backend & APIs:** Spring Boot, Spring Data, Node.js, Express.js, REST APIs, WebSockets, Server-Sent Events, JWT
- **Frontend & Mobile:** React, Redux, TailwindCSS, Capacitor (Android), Firebase Cloud Messaging
- **Systems & Reliability:** Job Queues, Worker Pools, Concurrency, At-Least-Once Delivery, Idempotency, Retry & Backoff, Rate Limiting, Pub/Sub
- **GenAI & Retrieval:** RAG, Hybrid Dense/Sparse Retrieval, Reciprocal Rank Fusion, Embeddings, Pinecone, Document Chunking
- **Data & Infrastructure:** PostgreSQL, MongoDB, Redis, Docker/Compose, Flyway, Prometheus, Grafana, Maven, Git/GitHub
- **Testing:** JUnit, Testcontainers, Integration Testing, GitHub Actions (CI)
- **AI Coursework:** LangChain, Model Context Protocol (MCP), Agentic Coding (Claude Code), Spec-Driven Development
- **Core Computer Science:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Object-Oriented Design

POSITIONS OF RESPONSIBILITY

- **General Associate, E-Cell NIT Silchar** (Aug 2024 – May 2026): led a **10-member** publicity team. **BD Intern, TrendyDice** (Jun – Aug 2025): conducted B2B outreach to **10+** campus clubs.